The new ClassX MOS solution.
Created with ease in mind to offer the better integration and the most powerful graphics to any Newsroom Computer System.

User Manual

rev. 24/03/2015

Stile ed Innovazione con ClassX

ClassX srl
Via Francesca 368/I
56020 S.Maria a Monte (PI) ITALY
Tel. Fax. ++39 0587 70 51 53
E-Mail: info@classx.it
Home-Page: www.classx.it

ORIGINAL EDITION © 2015

No part of this manual may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying and recording, for any purpose without the express written permission of a duly authorized representative of ClassX srl.

The information contained within this manual is believed to be true and correct at the time of publication.

ClassX, CastaliaCG, MoreCG, CoralCG, CGPlayout, LiveBoard, SGL, TemplateActions, LiveStadium, FrameMarker, MediaPlug are registered trademarks of ClassX srl.

All other brand names are trademarks of their respective holders.
All rights reserved. Unpublished rights reserved under International copyright laws.

1 Introduction

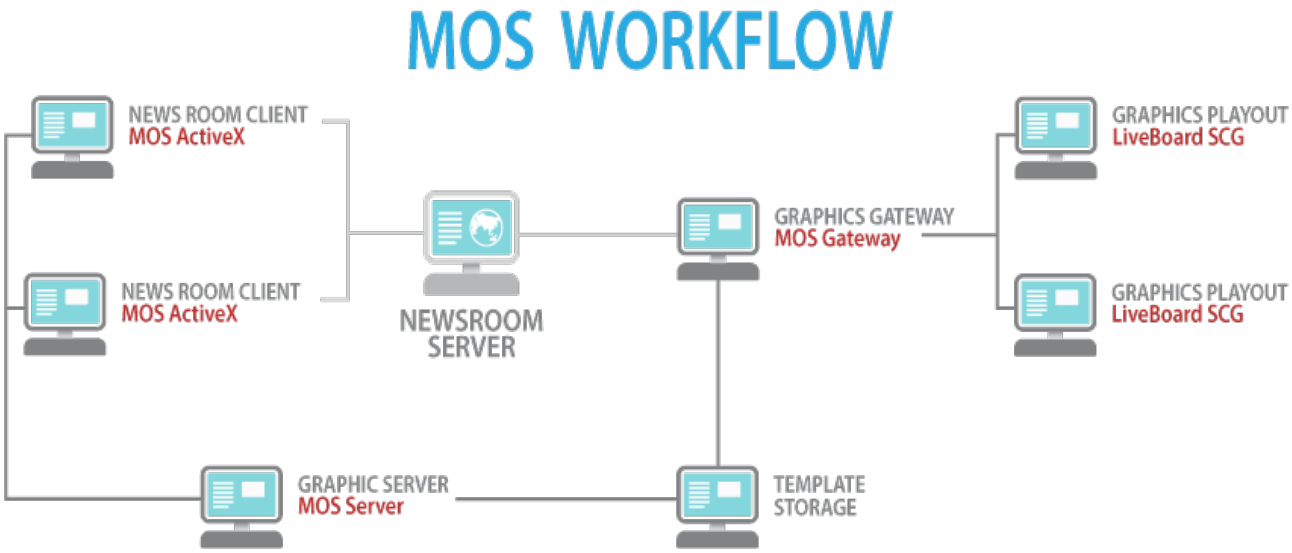
Congratulations for having purchased ClassX MOS Solution, the complete MOS solution for the integration with the most popular Newsroom Computer Systems.

1.1 What is the ClassX MOS Solution?

ClassX MOS Solution is a complete suite composed of two software (MOSServer e MOSGateway) and a plugin (ActiveX) which can be combined to offer a complete integration with the most popular NCS.
MOSGateway is the software that really implement the **MOS** protocol and is capable to handle the communication between the newsroom and the ClassX graphics playout software (LiveBoard with StealthCG). It provides the means to handle and select the graphics to play, taken from the incoming rundowns.

MOSServer and **ActiveX plugin** work together to allow the journalists to integrate the ClassX's graphics in their stories and rundowns: MOSServer loads on demand any graphics template created with MoreCG, CastaliaCG or CoralCG providing a real time preview to the ActiveX with the loaded template information. The ActiveX at this point allows the customization of the editable objects of the loaded template.

The scheme in figure clearly describes the architecture that links the ClassX MOS Solution.



MOSServer can be connected to more ActiveX running on different computers. Every ActiveX handle and displays the preview of the templates loaded in the MOSServer. The newsroom system will take care to communicate those information to the MOSGateway through the MOS protocol.

2 Installation

2.1 System Requirements

ClassX applications are subjected to a long and thorough beta-testing period in order to prevent malfunctions and ensure, at all times, the best performance. The information listed below will guide you in the process of setting up your workstation.

Supported Video Card:

Brand	Model
BlackMagic	DeckLink Studio/2
BlackMagic	DeckLink HD Extreme 3D
BlackMagic	DeckLink Pro
BlueFish444	SD/HD
Dayang	RedBridge II/III
Matrox	DSX/X.mio
AJA	Kona 3
NewTek	Tricaster

Supported VGA :

Brand	Model
NVidia	Any models NVidia DDR3 1GB or hig.

System Requirements:

Hardware/Software	Minimum	Recommended
Operative System	Windows XP Pro SP3	Windows 7 Pro SP1
Architecture	X32 for SD needs	X64 for HD needs
RAM	4GB DDR3	8GB DDR3
VGA	NVidia DDR3 1GB	NVidia 400 Series DDR5 1GB

2.2 System Configuration

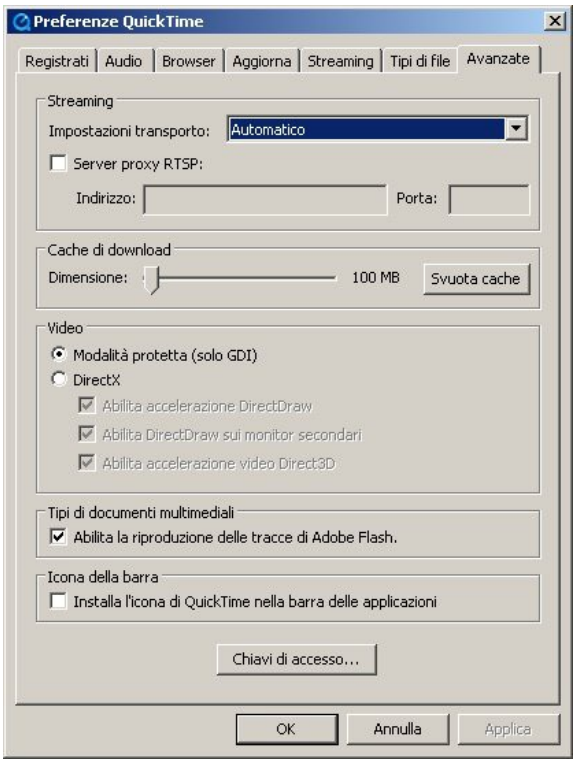
Windows Settings

- Disable **screensaver** (Control Panel/Display Properties /Screensaver/None).
- Disable **indexing files** (for all physical disks in My Computer/Right click on the icon of the disk/disable "List disk for fast file searching").
- Disable **Power Management** (Control Panel/Power Options).
- If your computer has at least 4GB ram to disable virtual memory (Right click "My Computer/Advanced/ Performance/Settings/Advanced/Virtual Memory/Change/No Page File").
- If the operating system is **Windows XP** set / 3GB switch in c: \ boot.ini:
  example:
  [Boot loader]
  timeout = 30
  default = multi (0) disk (0) rdisk (0) partition (1) \ WINDOWS
  [Operating systems]
  multi (0) disk (0) rdisk (0) partition (1) \ WINDOWS = "Microsoft Windows XP Home Edition" / noexecute = optin / fastdetect / 3GB

- If the operating system is **Windows 7,** you must disable Areo Peek.

QuickTime Settings

Start QuikTime Player, and select from menu "Edit" the "Preferences". Press then "**QuickTime Preferences**".



Media Settings / Audio / Video

- **Install FFDShow** 32 and/or 64-bit (install both if you use a 64bit system) for the decoding media files of popular formats. Then disable via "FFDShow Video decoder configuration", the codecs of the following formats: svq1, SVQ3, 8BPS, QTRLE, QTRPZA.
- **Windows 7** not allows the use of DirectShow filters for decode video files. This means that you can not use FFDShow filters if you want, for example, play the video in DivX/XviD format. To remove this limitation, we recommend the installation of Win7DSFilterTweaker that allows you to change the assignment of the codec.
- QuickTime does not support 64-bit architecture. It's not available for 64-Bit systems like Windows 7 64Bit.
- Install **QTFilter** present here for encoding the video in mov.

VGA output

- Connect, when the computer turned off, the monitor to the graphics card based on the available outputs: in most cases it is an S-Video, DVI or composite. To work properly you need the cables supplied with the card.
- Open the Display Properties and select the settings panel. If your device is connected and it was detected, the panel below shows two monitors: the number 1 is your primary monitor, the number 2 on the secondary outputs DVI / S-Video. Depending on the type of device you must configure resolution and refresh rate. Suppose you have connected the S-Video output to a television: in this case you need to set the resolution to 720x576 (PAL) and refresh rate of 50 Hz interlaced. For more information, visit the website of manufacturer's graphics card.

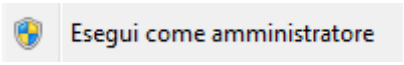
The information contained in this subsection are also available on our website at http://www.classx.it/en/support-en/software/system-requirements-en.html, which lists the recommended drivers for video cards supported.

2.3 Protection

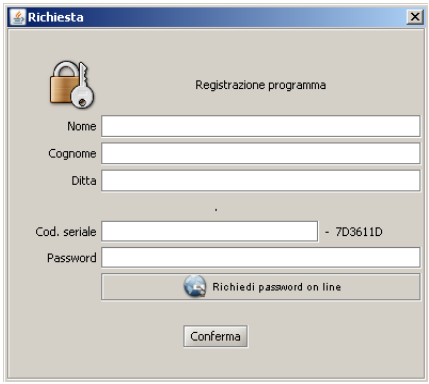
The biggest danger to your and our investment is the software piracy (since only that the hardware is recognized by current morality as the "theft"). Well, despite the time spent looking for an effective and painless protection for your SocialServer, someone copies oblivious the sanctions.
Recall that the time spent in the development of a new protection scheme has NOT been dedicated to the development of a new function or feature to the program. Recall also that the development of a new protection involves the writing of these amenities and obviously a greater hardship for registered users, who will be forced awkward maneuvers to secure their interests.

2.3.1 Entering the registration key

Once installed and configured the system, start SocialServer as Administrator to perform the registration. This requirement is essential for SocialServer, because only administrators can install software or hardware and access to all files on your computer. If you use Windows 7, press the right mouse on SocialServer icon, and select from the drop-down list:



Next, enter Name, Last Name, Company, Serial Number and Password to access into SocialServer, in the registration window. After you have typed all the data, just press "Submit" to start using the program. If the serial number and password have been entered correctly, the registration window will never be displayed, because the information will be part of the program. In this case, you can use SocialServer also by users who haven't administrator privileges.
The serial number and the password uniquely identifies the copy owned by the user that can use the program without any limitation (**WARNING: When inserting the passwords, observe the uppercase and lowercase**).
Remember that the serial number of the product is printed ouside the box.
To get your password you must send an email to ClassX specifying the personal data, serial number and the KeyID of the dongle attached.



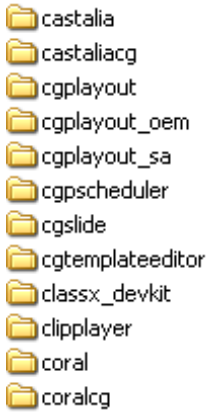
Notes: If you enter an incorrect password, SocialServer works in demo mode.

Warning: The serial number and password uniquely identify the user of the program. By installing the program, you agree not to disclose his personal key in any way.

Warning: The user is solely responsible for the safekeeping of the dongle. The key must be stored carefully and kept in a safe place after use. The loss of the key causes the termination of the license. To reactivate the software will have to buy a new license. If the key to be defective, ClassX agrees to replace free of charge. The key remains the property of ClassX and is granted solely to the user's use of the software ClassX.

2.4 Organization of files and directories

After the SocialServer installation process, on our hard drive we will find a folder called "ClassX_Applications" containing files and folders for the functioning of the application.
In detail, the folder "ClassX_Applications" contains the following directories:



Folder "SocialServer" - contains the application and everything you need SocialServer.
Folder "jre" - contains the directories and libraries needed by the ClassX application.
"Shared" folder - contains all the resources shared by ClassX applications (Images, Video, DataSource, Scripts, Audio, 3D objects, preset themes, etc..).

The folder "SocialServer" contains the main program, its support files and any additional trays to complete the program. The following are the contents of SocialServer directory:
Folder "doc" - contains all the documents in HTML format, for the manual SocialServer.
Folder "projects" - contains all the projects that come with SocialServer.
Folder "export" - contains all the animations and sequences of frames generated by SocialServer.
Folder "presets" - contains all the presets that come with various versions of SocialServer.

With the introduction of 64-bit architecture, folder "SocialServer" contains two executables: SocialServer64.exe and SocialServer.exe
As can be easily understand, in the name of the executable that has the suffix 64, will start the 64-bit version, while the other will work in 32 bit.
The file name or SocialServer_c.exe SocialServer64_c.exe allows you to start SocialServer using console.

Warning: The folder "platform", "shared / plugins" contains files necessary to run SocialServer. These drawers and related files should not be neither moved, renamed or altered.

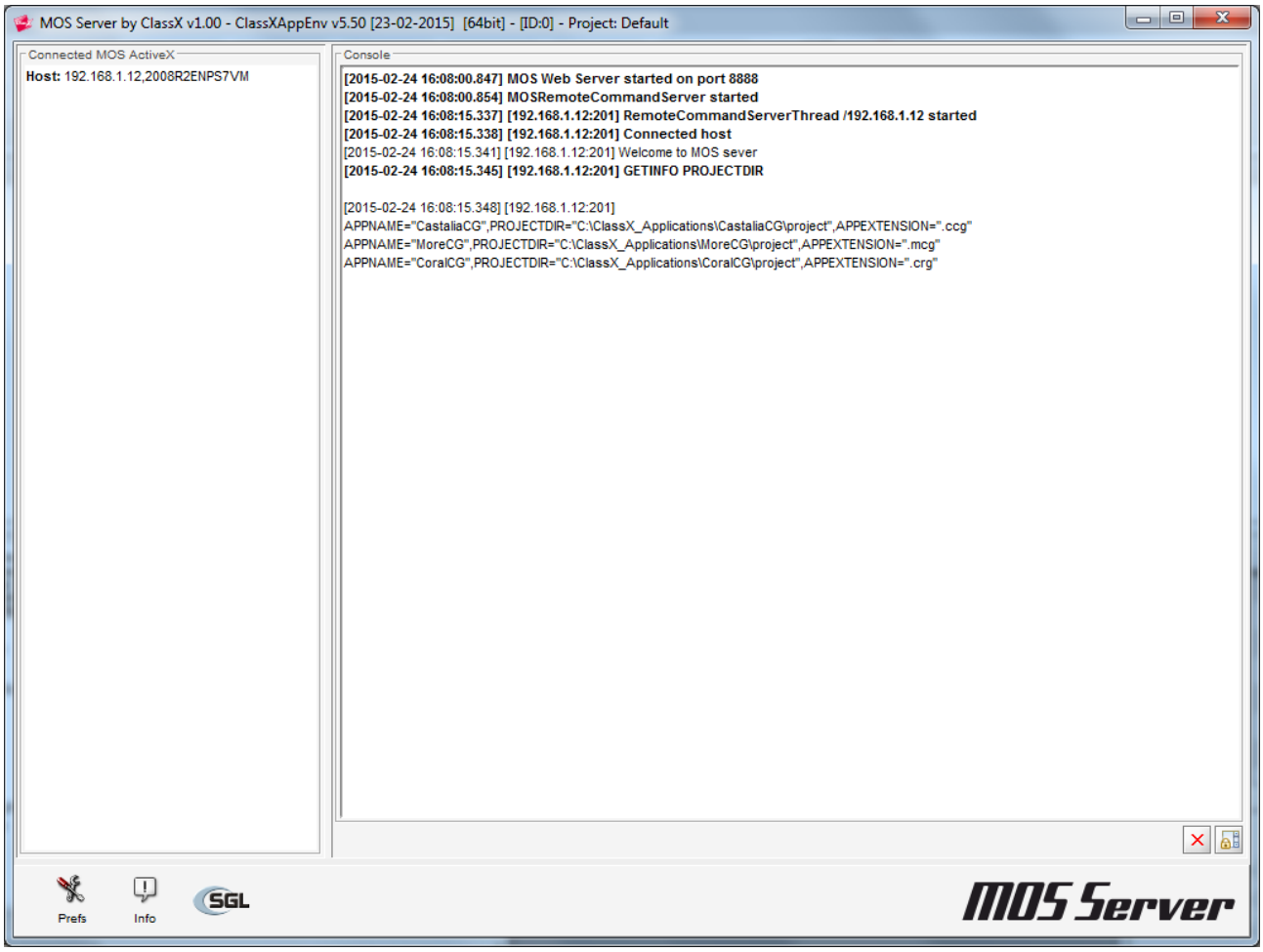
2.4.1 The conventions for the name of the files

To further facilitate the work of maintaining on-disk file is advisable to use appropriate names, chosen on the basis of small and healthy everyday rules that can greatly reduce search times.
For example, if we were to save a project SocialServer to a topic, it would be advisable to use the name "Argument.cst" and the corresponding sequence of frames with "Argument_xxx.jpg".
The name "topic" would immediately understand what it is, while suffixes ".cst" and ".jpg" serve to indicate the file type.
In general, therefore, the prefix should indicate the category of file (eg "Sigle_", "Tit_" etc..), While the suffix represents the type (ex ".gif", ". jpg", ". cst ", etc...)

Note: SocialServer automatically adds the appropriate extensions when saving files.

3 MOSServer

MOSServer is the handler of the graphics contents created with MoreCG, CastaliaCG and CoralCG and it has been designed with a minimal interface that is extremely easy to use.
As shown in the underlying figure, the MOSServer interface is divided in two sections: the left section contains the list of the connected ActiveX (the journalists), and the right section contains a text area that logs all the messages exchanged between the MOSActiveX and the MOSServer.

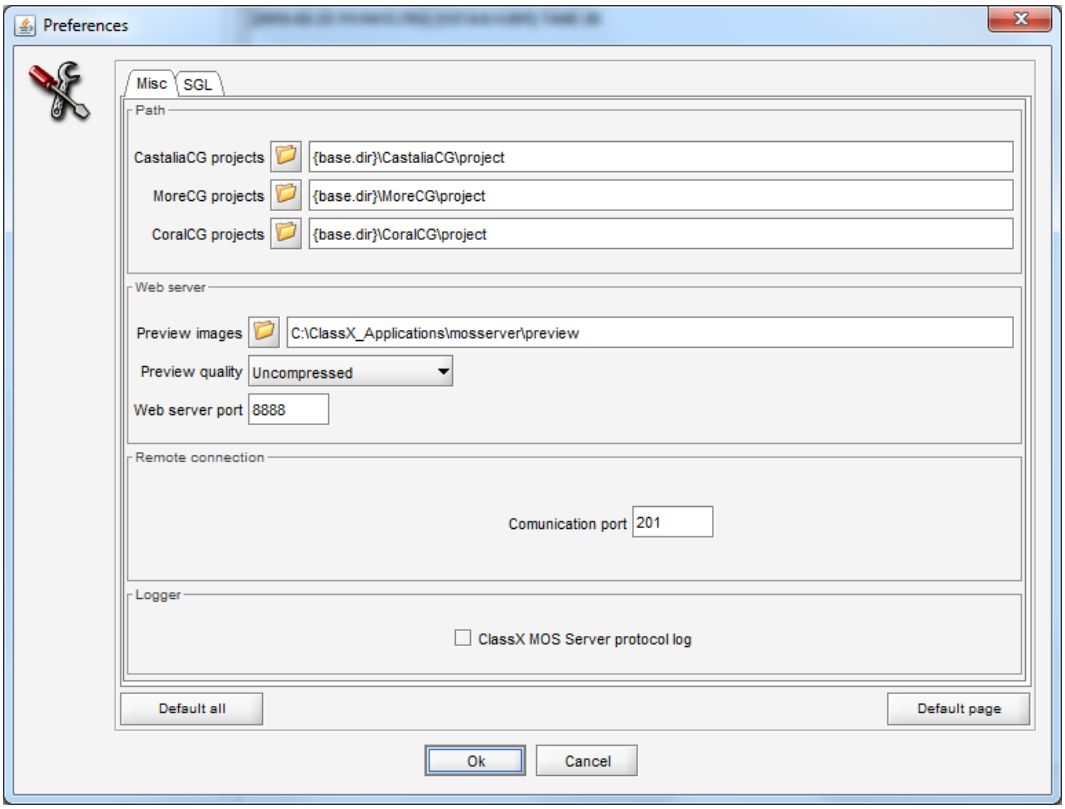


3.1 The ToolBar

The Toolbar provides access to application preferences, PDF documentation and program information.

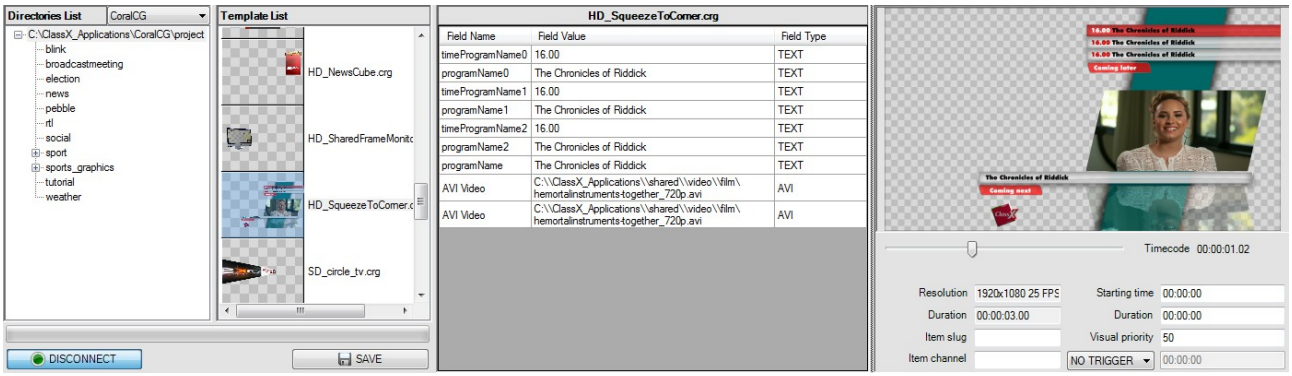
3.2 MOSServer preferences

MOSServer can handle any ClassX template, but it needs to know the paths of directories which contain the templates.
MOSServer, furthermore, integrates a simple web server used to display the preview images of items inserted in the Stories by the ActiveX. In this panel the user can also choose the quality of the images and the tcp/ip port where the web server will answer.



4 MOSActiveX

MOSActiveX is an ActiveX plugin that integrates itself seamlessly with MOS-compliant NCS that support this technology. MOSActiveX can show to the journalist which template he's editing as well as all the editable objects and a real time preview. The user interface can be divided into three main sections: the list of directories and templates, the table with all the editable objects and the real time preview area which also contains relevant information about the loaded template.



Once the connection has been established with the MOSServer, the user can choose the desired directory and the desired template from the list. After the template has been selected, he can start to edit the template fields. The ActiveX will always display a real time preview with the actual values inserted by the journalist.

As soon as the edit is terminated, it will be possible to insert the item in the story by clicking the button “SAVE”; if more changes are needed after the item has been saved in the story, the journalist can start the edit again by double-clicking the desired item. At this point it will be possible to save it again in the story by clicking the button “SAVE”.

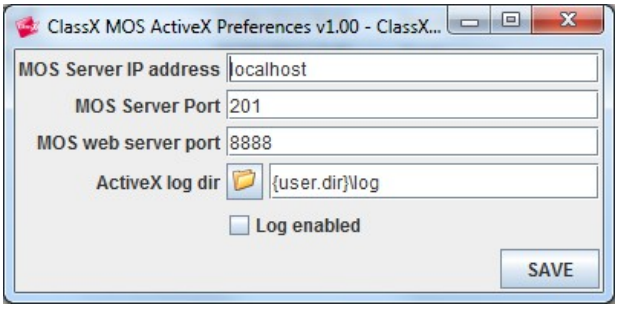
4.1 Configuring the MOSActiveX

Installing the ActiveX is a simple operation, but it requires some steps:

- Be sure that the ClassX Applications have been correctly installed.
- Be sure that the .NET Framework is installed and up to date (the ActiveX requires at least the 4.0 version).
- Run the “install.bat” in the folder “ClassX_Applications\mosactivex\activex”.
- At this point, if everything went fine, the ActiveX should be installed and registered, with *ClassXMOSActiveX.InteropUserControl* as entry point in the Windows registry.

To complete the configuration is necessary to configure the preferences of the ActiveX to allow it to connect with MOSServer.

This can be done with the software installed in the folder “ClassX_Applications\mosactivex”. Those preferences are saved in the Windows registry and loaded by the ActiveX at every start up.



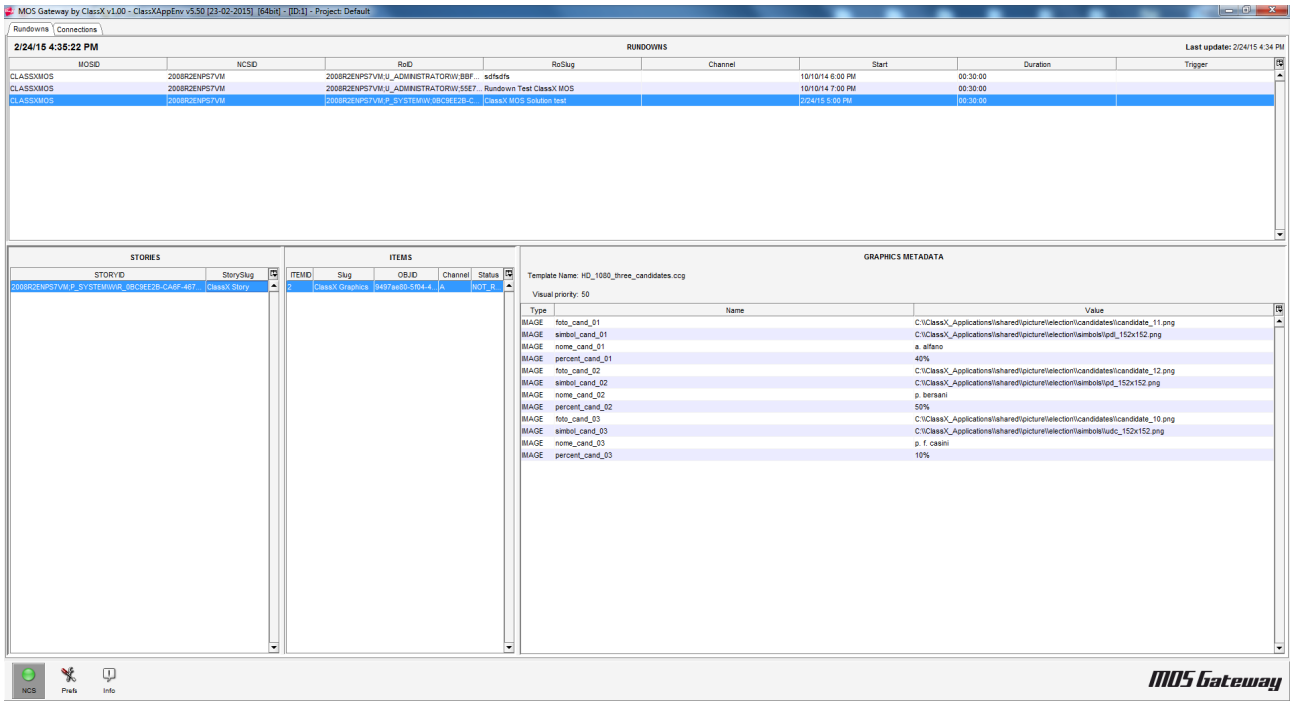
Please note. Windows, in the 64 bit version of the system, maintains two different instances of the registry. By default it is recommended to use the 32 bit version of the ActiveX prefs editor. In case the ActiveX cannot read the saved values the user can use the 64 bit version of the prefs editor.

5 MOSGateway

MOSGateway is the software that handles the connection between the newsroom system and the ClassX playout, accepting MOS commands and converting them in remote commands for the playout.

MOSGateway has been designed with a minimal user interface which makes it very simple but effective: the user interface offers two main tabs: the first that contains the incoming rundowns and the second that contains the connections and log information.

The first tab is divided into two main sections: the upper area that contains the rundown list and the lower area that contains all the information of the selected rundown.
In these areas it is possible to manage rundowns and play the relative graphics both in **auto mode** or in **manual mode**, according to the kind of trigger set in the rundown.



The other tab instead contains important information about the connections with the newsroom and the playout. In the upper area: at left we can find information about the connections with the newsroom system. At right we can find information about the connections with the ClassX Graphics playout.

The lower area of this panel contains the log with all the information about the exchanged messages between MOSGateway, newsroom system and the graphics playout, besides all the information about eventual problems and errors. In order to work properly, the log must be configured in the preferences panel of the MOSGateway and it needs a MySQL database to store all the information.

The log panel contains also some useful tools as the filter and the export button.

5.1 Configuring the MOSGateway

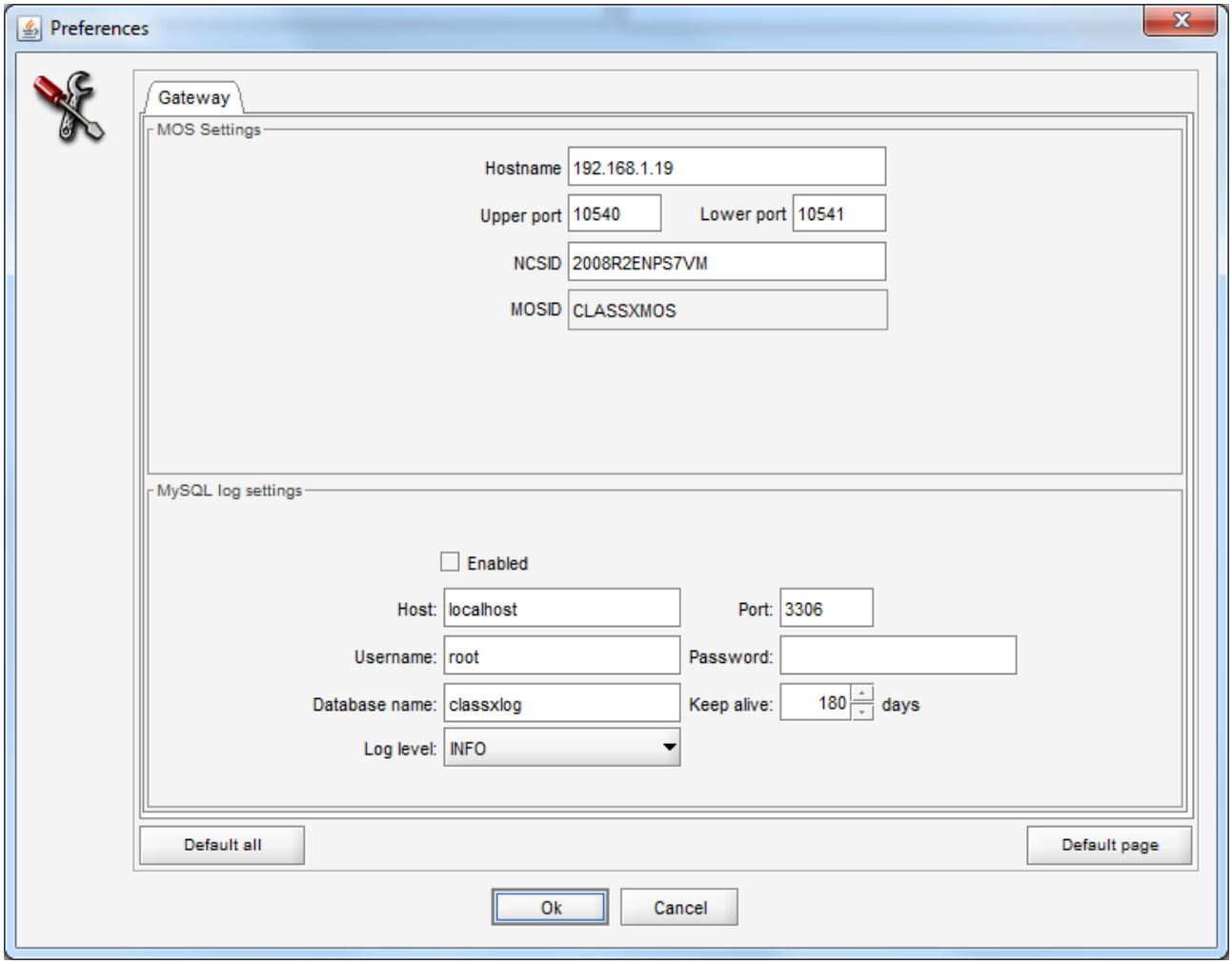
The MOSGateway preferences contains both the MOS settings and the MySQL log settings.

The MOS settings are:

- Hostname: the hostname/address of the NCS.
- Upper and Lower port: ports of the NCS.
- NCSID: Identifier of the NCS (usually the machine name).

The mySQL log settings are:

- Host/Port/Username/Password: connections info of the MySQL database.
- Database name: the name of the db which will contains the log. It needs to be manually created.
- Keep alive: records older than this value will be deleted from the log.
- Log level: maximum log level; log messages with a finer level will be ignored.



5.2 The ToolBar

The Toolbar provides the access to application preferences, PDF documentation and program information.

6 For the users

6.1 Programming errors

ClassX MOS Solution is an innovative software suite not only for the user interface, fast and powerful, but also for its versatility: it is perfectly compatible with Windows 2000, Windows XP, Windows Vista, Windows 7.
ClassX MOS Solution was developed with the utmost care and has been tested on different configurations for a long time.
The program should be free from problems of illegal access to memory problems and release of resources.
However, being developed and tested by humans, could suffer from problems never encountered during the tests, so please anyone found any problems, send us (by letter or fax or mail) a message about it.
Because the message is as useful as possible, for the purpose of solving a problem, it should be specified:

- Computer model.
- Version of the OS.
- Maximum amount of memory.
- Type of graphics card.
- External hardware devices.
- Version of ClassX MOS Solution (see the title bar).
- Accurate verbal description of the sequence of operations that led to the error.
- A copy of the contents of the panel "Information".

6.2 Collaboration

Do you have any new ideas for improvement ClassX MOS Solution?
Do you have advice and/or constructive criticism?
You want to develop a new effect or a new form of Export?
Any such communication will be seriously considered and evaluated for the development in a next version of ClassX MOS Solution.

6.3 How to obtain support via internet by ClassX

For all users who have Internet access, a free service is available to support the program. For any questions or suggestions, please send an email to **info@classx.it** remembering to include in the subject line the word "SUPPORT ClassX MOS Solution".
All the latest news of ClassX MOS Solution and about other products can be found at ClassX **WWW.CLASSX.IT**

6.4 Important

Nothing contained in this document will give rise to representations or warranties, express or implied, with respect to the products described herein, the functionality, compatibility and availability.
In addition, the ClassX assumes no responsibility or liability for statements or assertions made by itself or by third parties or vendors in the prints reproduced here.
In any case, the ClassX be liable for any direct, indirect, incidental or consequential damages resulting from any claims that may arise from allegations here, although there was no information on the possibility of such damages.

Index

1 INTRODUCTION **4**
1.1 What is the ClassX MOS Solution? 4

2 INSTALLATION **5**
2.1 System Requirements 5
2.2 System Configuration 5
2.3 Protection 7
2.3.1 Entering the registration key *7*
2.4 Organization of files and directories 8
2.4.1 The conventions for the name of the files *8*

3 MOSSERVER **9**
3.1 The ToolBar 9
3.2 MOSServer preferences 10

4 MOSACTIVEX **11**
4.1 Configuring the MOSActiveX 11

5 MOSGATEWAY **12**
5.1 Configuring the MOSGateway 13
5.2 The ToolBar 13

6 FOR THE USERS **14**
6.1 Programming errors 14
6.2 Collaboration 14
6.3 How to obtain support via internet by ClassX 14
6.4 Important 14